Science · IX · Ch. 11 · Educart

cl9ch11 — extracted questions & answers

71 top-level · 144 logical questions · 100 with an answer in the document · 21 explicitly not in the document

1. Example 1. Case Based:
Any object that vibrates makes sound. A medium is the matter or substance through which sound is transferred. It can take the form of a solid, a liquid, or a gas. From the place of generation to the listener, sound travels through a medium. When an object vibrates, it causes the medium's particles to vibrate as well. The particles do not make it all the way to the ear from the vibrating object. Mechanical waves are sound waves that are defined by the motion of particles in the medium. When a vibrating object goes ahead, it compresses and pushes the air in front of it, generating a high-pressure zone known as a compression (C) zone. When a vibrating object goes backwards, it forms a low-pressure zone known as rarefaction. Therefore, the sound is a longitudinal wave.

p. 316 · case_study · worked_example · blocks: blk_000044, blk_000045

A. (A) A full variation in density/pressure of a medium from maximum to minimum and back to maximum owing to longitudinal wave propagation is known as:
(a) frequency
(b) pitch
(c) amplitude
(d) oscillation
(Remember)

p. 316 · mcq · worked_example · blocks: blk_000046

Answer: Ans. (A) (d) oscillation
Explanation: A longitudinal wave (such as a sound wave) consists of compressions (high-pressure regions) and rarefactions (low-pressure regions). A full variation from maximum density/pressure to minimum and back to maximum constitutes one complete oscillation.

source blocks: blk_000051

B. (B) When a sound wave goes from air into water, which of the following quantity remains unchanged?
(a) Velocity
(b) Intensity
(c) Wavelength
(d) Frequency
(Understand)

p. 316 · mcq · worked_example · blocks: blk_000047

Answer: (B) (d) Frequency
Explanation: When a sound wave travels from air into water, the frequency does not change as it is determined by the source of the wave.

source blocks: blk_000052

C. (C) Assertion (A): Pressure waves are the longitudinal waves.
Reason (R): When compression and rarefaction are created, longitudinal waves propagate in a medium involving changes in pressure and volume of air.
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true. (Understand)

p. 316 · assertion_reason · worked_example · blocks: blk_000048

Answer: (C) (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
Explanation: Pressure waves are longitudinal waves because they propagate through compressions (high pressure) and rarefactions (low pressure). This occurs due to changes in pressure and volume in the medium, allowing the wave to travel. Since the reason correctly explains this process, both the assertion and reason are

source blocks: blk_000053

D. (D) The speed of sound is finite and is much less than the speed of light and it depends on some factors. Can you list down those factors?
(Understand)

p. 316 · conceptual · worked_example · blocks: blk_000049

Answer: (D) The speed of sound depends upon:
(1) The properties (elasticity and density) of the medium through which it propagates.
(2) The temperature of the medium. Higher temperature increases the speed of sound in gases.
(3) Medium: Sound travels fastest in solids, slower in liquids, and slowest in gases due to particle density.
(Any two)

source blocks: blk_000054

E. (E) What type of wave is represented by:
(i) Density-distance graph?
(ii) Displacement-distance graph?
(Understand)

p. 316 · conceptual · worked_example · blocks: blk_000050

Answer: (E) (i) Density-distance graph represents Longitudinal waves. These waves propagate through a medium by creating alternating compressions and rarefactions. Compression is a region where air particles are close together, leading to higher density and pressure than normal air. Rarefaction is a region where air particles are farther apart, resulting in lower density and pressure than normal air.
On a

source blocks: blk_000056

2. Example 2. Explain how sound is produced by your school bell.
(Apply) [NCERT]

p. 317 · conceptual · worked_example · blocks: blk_000069

Answer: Ans. The sound generated by the school bell depends on its vibrations when struck. When the bell is struck physically or mechanically, it causes the surrounding air particles to vibrate, creating a disturbance in the medium. As the bell moves forward, it compresses the air particles, forming a high-pressure region (compression). When it moves backward, it creates a low-pressure region (rarefaction

source blocks: blk_000069

3. Example 3. Suppose you and your friend are on the moon. Will you be able to hear any sound produced by your friend?
(Apply) [NCERT]

p. 317 · conceptual · worked_example · blocks: blk_000070

Answer: Ans. Moon has no atmosphere i.e., it is a vacuum. Since sound waves need a material medium like air to travel, they cannot propagate in a vacuum. Therefore, I would not be able to hear any sound produced by my friend on the moon.

source blocks: blk_000070

4. Example 4. Which wave property determines
(A) loudness (B) pitch? (Understand) [NCERT]

p. 320 · conceptual · worked_example · blocks: blk_000127

Answer: Ans. (A) The amplitude of a sound wave determines its loudness. A greater amplitude results in a louder sound.
(B) The frequency of the wave determines its pitch. A higher frequency corresponds to a higher-pitched sound.

source blocks: blk_000128

5. Example 5. Guess which sound has a higher pitch: guitar or car horn. (Apply) [NCERT]

p. 320 · conceptual · worked_example · blocks: blk_000129

Answer: Ans. Pitch depends on the frequency of vibration. A sound with a higher frequency is perceived as having a higher pitch. Since the guitar produces vibrations at a higher frequency than a car horn, it has a higher pitch.

source blocks: blk_000130

6. Example 6. Calculate the wavelength of a sound wave whose frequency is 220 Hz and speed is 440 m/s in a given medium. (Apply) [NCERT]

p. 320 · numerical · worked_example · blocks: blk_000134

Answer: Ans. Given,
Frequency of the sound wave, $\nu = 220$ Hz
Speed of the sound wave, $v = 440$ m/s
Speed = Wavelength × Frequency
$v = \lambda \times \nu$
$440 = \lambda \times 220$
$\lambda = 440 / 220$
$= 2$ m
Hence, the wavelength of the sound wave is 2 m.

source blocks: blk_000135

7. Example 7. A person is listening to a tone of 500 Hz sitting at a distance of 450 m from the source of the sound. What is the time interval between successive compressions from the source? (Apply) [NCERT]

p. 320 · numerical · worked_example · blocks: blk_000136

Answer: Ans. Frequency of waves, $\nu = 500$ Hz
The time period of a wave is defined as the interval between two subsequent compressions or rarefactions, and it is given by,
$T = 1 / \nu$
$= 1 / 500 = 0.002$ s
$T = 2$ ms (milli seconds)
Hence, the period of sound waves is equal to the time interval between successive compressions from the source of the sound.

source blocks: blk_000137

8. Example 8. Distinguish between loudness and intensity of sound. (Analyse) [NCERT]

p. 320 · distinguish · worked_example · blocks: blk_000138

(3) The ear's response to sound is measured by its loudness. Loudness is a personal preference.	(3) The sound power per unit area is known as intensity.
(4) It is measured in decibels (dB).	(4) It is measured in watts per square meter (W/m^2).
(5) It is influenced by the sensitivity of human ears.	(5) It is not affected by human perception.
(6) It is a subjective quantity. A person's hearing can be loud for one person but not for another.	(6) It is an objective quantity. It doesn't differ from person to person.

Answer: Ans. Loudness | Intensity of Sound
(1) It measures the ear's response to sound. | (1) It measures the power of a sound wave per unit area.
(2) It depends on amplitude and is perceived by the listener. | (2) It depends on amplitude but is independent of human perception. (3) The ear's response to sound is measured by its loudness. Loudness is a personal preference. | (3) The sound power per unit ar

source blocks: blk_000139, blk_000140, blk_000142

9. Example 9. Why are the ceilings of concert halls curved? (Apply) [NCERT]

p. 323 · conceptual · worked_example · blocks: blk_000187

Answer: Ans. The ceilings of concert halls, cinema halls, and conference halls are curved to ensure that sound reaches all corners of the space uniformly after reflection from the curved surface.

source blocks: blk_000187

10. Example 10. Case Based:
Sushant, a ninth-grade student, once fell ill with a cold, cough, and high fever. His father took him to the doctor, who examined his chest and back using a stethoscope. During the examination, the doctor asked Sushant to take deep breaths while listening through the stethoscope. After assessing his condition, the doctor prescribed medication. By taking the prescribed medicines regularly, Sushant recovered within four days. However, he was curious about why the doctor used a stethoscope for his checkup. During the winter break, his cousin Karan, a medical student, visited their home. At Sushant's request, Karan explained the function and working of a stethoscope.

p. 323, 324 · case_study · worked_example · blocks: blk_000189, blk_000190

A. (A) SI unit of sound is:
(a) Metre
(b) Hertz
(c) Decibel
(d) Metre/second
(Remember)

p. 323 · mcq · worked_example · blocks: blk_000192

Answer: Ans. (A) (c) Decibel
Explanation: SI unit of sound is decibel (dB). This unit is mostly used to describe the ratio of one physical property's value to another on a logarithmic scale. The Decibel scale, or DB scale, is used for this. Furthermore, the unit can be used to represent a variety of reference pressures and intensities.

source blocks: blk_000200

B. (B) The doctor heard the sound of Sushant's internal organs with the help of stethoscope. Do you know, on what principle does the stethoscope works?
(a) Echo
(b) Infrasonic

(c) Reflection
(d) Multiple reflections
(Understand)

p. 323 · mcq · worked_example · blocks: blk_000193

Answer: (B) (d) Multiple reflection
Explanation: A stethoscope works on the principle of multiple reflections of sound. The sound from internal organs is received by the chest piece and travels through the tube, reflecting multiple times before reaching the doctor's ears. This amplification helps the doctor hear internal body sounds clearly.

source blocks: blk_000202

C. (C) Assertion (A): When sound collides with a hard, polished surface, an echo is created.
Reason (R): Objects with a delicate and loose texture can completely reflect sound energy.

p. 323, 324 · assertion_reason · worked_example · blocks: blk_000194

a. (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true. (Understand)

p. 324 · mcq · worked_example · blocks: blk_000196

Answer: (C) (c) (A) is true but (R) is false.
Explanation: Echo occur when sound waves hit a hard, smooth surface and reflect back. However, objects with a soft or loose texture, like curtains or carpets, absorb sound instead of reflecting it, reducing echo.

source blocks: blk_000203

D. (D) What is the purpose of using a stethoscope?
(Remember)

p. 324 · conceptual · worked_example · blocks: blk_000197

Answer: (D) The stethoscope is medical equipment that is used by doctors to listen to internal body sounds, such as heartbeat, lung sounds, and blood flow. It helps in diagnosing various medical conditions related to the heart and lungs.

source blocks: blk_000204

E. (E) Explain loudness of sound. On what factors does it depend on? (Understand)

p. 324 · conceptual · worked_example · blocks: blk_000198

Answer: (E) Loudness of a sound is a subjective quantity and a measure of sound energy that reaches the ear per second. It depends upon:
(1) Amplitude of the vibrating body.
(2) Sensitivity of the human ear.

source blocks: blk_000205

1. 1. Before playing the orchestra in a musical concert, a sitarist tries to adjust the tension and pluck the string suitably. By doing so, he is adjusting:
(a) intensity of sound only
(b) amplitude of sound only
(c) frequency of the sitar string with the frequency of other musical instruments
(d) loudness of sound [NCERT Exemplar]

p. 327 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000252

Answer: Ans. (c) frequency of the sitar string with the frequency of other musical instruments Explanation: Before playing in an orchestra, a sitarist adjusts the tension and plucks the string to match the frequency of the sitar with other musical instruments. This process is known as tuning.

source blocks: blk_000253, blk_000254

2. 2. Most television sets these days can be operated through a Remote Control. How do most 'remotes' communicate with TV sets?
(a) Using radio waves
(b) Using infrared rays
(c) Using ultraviolet rays
(d) Using microwaves [Delhi Gov. QB 2022]

p. 327 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000256

Answer: Ans. (b) Using infrared rays Explanation: TV remote controls use infrared (IR) rays to communicate with television sets. The remote control has an LED present in it that emits pulses of infrared light in a specific pattern corresponding to different commands such as volume up, channel change, or power on/off. The TV's IR sensor (receiver) detects these signals and interprets them to perform the de

source blocks: blk_000257, blk_000258

3. 3. Sound travels in air if:

p. 327 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000259

a. (a) particles of medium travel from one place to another.

p. 327 · mcq · solved_question · blocks: blk_000259

Answer: Ans. (c) disturbance moves.

source blocks: blk_000260

b. (b) there is no moisture in the atmosphere.

p. 327 · mcq · solved_question · blocks: blk_000259

Answer: Ans. (c) disturbance moves.

source blocks: blk_000260

c. (c) disturbance moves.

p. 327 · mcq · solved_question · blocks: blk_000259

Answer: Ans. (c) disturbance moves.

source blocks: blk_000260

d. (d) both particles as well as disturbance travel from one place to another.
[NCERT Exemplar]

p. 327 · mcq · solved_question · blocks: blk_000259

Answer: Ans. (c) disturbance moves.

source blocks: blk_000260

4. 4. The speed of sound waves in air:
(a) is independent of humidity
(b) is independent of temperature
(c) increases with increase in humidity
(d) decreases with increase in humidity.

[DIKSHA]

p. 327 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000262

Answer: Ans. (c) increases with increase in humidity

source blocks: blk_000263

5. 5. If the speed of the wave is 120 m/s and its frequency is 2000 Hz, then wavelength for this wave in cm will be:
(a) 6
(b) 0.6
(c) 60
(d) 600

p. 327 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000265

Answer: Ans. (a) 6

source blocks: blk_000266

6. 6. In which of these conditions will an echo be heard?

p. 327 · conceptual · solved_question · 1 mark (section_banner) · blocks: blk_000271

a. (a) A man playing drums on a beach.

p. 327 · mcq · solved_question · blocks: blk_000271

Answer: Ans. (d) A man shouting from a place that is surrounded by hills.

source blocks: blk_000273

b. (b) A man reciting poems in a small room.

p. 327 · mcq · solved_question · blocks: blk_000271

Answer: Ans. (b) 540 m

source blocks: blk_000288

c. (c) A man shouting from the middle of an open farm.

p. 327 · mcq · solved_question · blocks: blk_000271

Answer: Ans. (d) A man shouting from a place that is surrounded by hills.

source blocks: blk_000273

d. (d) A man shouting from a place that is surrounded by hills.
[CBSE Question Bank 2024]

p. 327 · mcq · solved_question · blocks: blk_000271

Answer: Ans. (d) A man shouting from a place that is surrounded by hills.

source blocks: blk_000273

7. 7. Two sound waves in air have a wavelength ratio 3 : 7. Then their frequency ratio will be:
(a) 3 : 7
(b) 2 : 7
(c) 7 : 3
(d) 1 : 1

p. 328 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000276

Answer: Ans. (c) 7 : 3

source blocks: blk_000277

8. 8. Crests and troughs are not formed in which type of waves:
(I) Transverse waves
(II) Longitudinal waves
(III) Electromagnetic waves
(IV) Sound waves
Options:
(a) (II) and (IV)
(b) Only (I) and (II)
(c) (I), (III) and (IV)
(d) (II) and (IV)

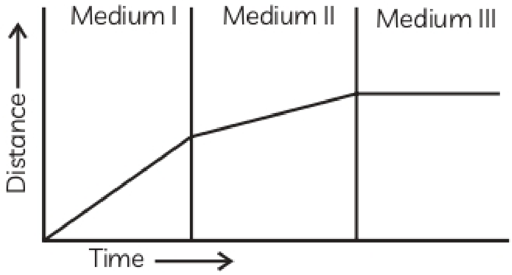
p. 328 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000279

Answer: Ans. (d) (II) and (IV)

source blocks: blk_000280

9. 9. The graph shows how the speed of a sound changes as it travels through medium I, then medium II and then medium III.
What are the media?
(a) Gas, liquid and solid
(b) Liquid, gas and solid
(c) Solid, gas and liquid
(d) Solid, liquid and gas

p. 328 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000283



Answer: Ans. (d) Solid, liquid and gas

source blocks: blk_000285

10. 10. A trawler's echo sounder hears an echo from a shoal of fish 0.6 seconds after it was sent. How deep is the shoal if the sound speed in water is 1800 m/s?
(a) 300 m
(b) 540 m
(c) 750 m
(d) 1080 m

p. 328 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000287

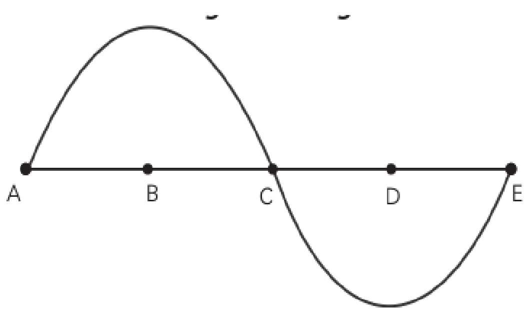
Answer: Ans. (b) 540 m

source blocks: blk_000288

11. 11. Half of the wavelength in the given curve is:

(a) AC
(b) BC
(c) BD
(d) DE

p. 329 · mcq · solved_question · 1 mark (section_banner) · blocks: blk_000293



Answer: Ans. (c) BD

source blocks: blk_000295

12. 12. Assertion (A): Every vibrating body generates sound.
Reason (R): All sounds are not audible.

p. 329 · assertion_reason · solved_question · 1 mark (section_banner) · blocks: blk_000301

Answer: Ans. (a) Both A and R are true and R is the correct explanation of A.

source blocks: blk_000302

13. 13. Assertion (A): On a hot summer day, sound travels faster than on a freezing winter day.
Reason (R): The square of the absolute temperature of sound is directly proportional to its velocity.

p. 329 · assertion_reason · solved_question · 1 mark (section_banner) · blocks: blk_000304

Answer: Ans. (c) (A) is true but (R) is false.

source blocks: blk_000305

14. 14. Assertion (A): A wave is a disturbance that travels through a medium when particles in the medium cause motion in their neighbours.
Reason (R): Carbon bonding allows the medium particles to go forward by themselves.

p. 329 · assertion_reason · solved_question · 1 mark (section_banner) · blocks: blk_000307

Answer: Ans. (c) (A) is true but (R) is false.

source blocks: blk_000308

15. 15. Assertion (A): The sound produced by the sitar is shriller than that produced by a tabla.
Reason (R): The sitar's frequency will be higher than the frequency produced by the tabla.

p. 329 · assertion_reason · solved_question · 1 mark (section_banner) · blocks: blk_000311

Answer: Ans. (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

source blocks: blk_000312

16. 16. Assertion (A): Longitudinal waves are produced by rotating the piston back and forth in a cylinder containing a liquid.
Reason (R): The particles of the medium oscillate parallel to the wave's propagation direction in longitudinal waves.

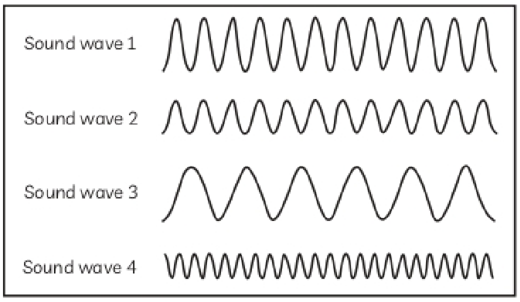
p. 330 · assertion_reason · solved_question · 1 mark (section_banner) · blocks: blk_000316

Answer: Ans. (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
Explanation: The particles of the medium oscillate back and forth parallel to the wave's propagation direction in a longitudinal wave.
When a cylinder's piston is moved back and forth fast, the piston is actually vibrating in a direction parallel to the cylinder's length. The liquid particles in the cylinder begin to vi

source blocks: blk_000317

17. 17. The picture shows four different sound waves. On the basis of given information, answer the following questions:

p. 330, 331 · case_study · solved_question · 4 marks (section_banner) · blocks: blk_000320, blk_000322



A. (A) Which sound wave has the highest frequency?
(a) Sound wave 1
(b) Sound wave 2
(c) Sound wave 3
(d) Sound wave 4

p. 330 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000323

Answer: Ans. (A) (d) Sound wave 4
Explanation: The sound wave with the highest frequency is sound wave 4 because it has the most compressed waves, indicating a higher frequency.
(B) (c) Sound wave 1 and sound wave 3
Explanation: The two sound waves that have almost the same loudness are sound wave 1 and sound wave 3, as they have similar amplitudes.
(C) (d) 35000 Hz
Explanation: The frequency of an ultras

source blocks: blk_000328

B. (B) Which two sound waves have almost the same loudness?
(a) Sound wave 1 and sound wave 2
(b) Sound wave 2 and sound wave 4
(c) Sound wave 1 and sound wave 3
(d) Sound wave 3 and sound wave 4

p. 330 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000324

Answer: Ans. (A) (d) Sound wave 4
Explanation: The sound wave with the highest frequency is sound wave 4 because it has the most compressed waves, indicating a higher frequency.
(B) (c) Sound wave 1 and sound wave 3
Explanation: The two sound waves that have almost the same loudness are sound wave 1 and sound wave 3, as they have similar amplitudes.
(C) (d) 35000 Hz
Explanation: The frequency of an ultras

source blocks: blk_000328

C. (C) Which of these is the frequency of an ultrasound?
(a) 10 Hz
(b) 75 Hz
(c) 15000 Hz
(d) 35000 Hz

p. 330 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000325

Answer: Ans. (A) (d) Sound wave 4
Explanation: The sound wave with the highest frequency is sound wave 4 because it has the most compressed waves, indicating a higher frequency.
(B) (c) Sound wave 1 and sound wave 3
Explanation: The two sound waves that have almost the same loudness are sound wave 1 and sound wave 3, as they have similar amplitudes.
(C) (d) 35000 Hz
Explanation: The frequency of an ultras

source blocks: blk_000328

D. (D) If the density of air at a point through which a sound wave is passing is at its maximum at a given instant, what will be the pressure at that point?
(a) Minimum
(b) Equal to the density of air
(c) Equal to the atmospheric pressure
(d) Maximum

p. 330 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000326

Answer: Ans. (A) (d) Sound wave 4
Explanation: The sound wave with the highest frequency is sound wave 4 because it has the most compressed waves, indicating a higher frequency.
(B) (c) Sound wave 1 and sound wave 3
Explanation: The two sound waves that have almost the same loudness are sound wave 1 and sound wave 3, as they have similar amplitudes.
(C) (d) 35000 Hz
Explanation: The frequency of an ultras

source blocks: blk_000328

D. OR
(D) If these sound waves were all produced at the same time, which would reach a listener first?
(a) Sound wave 1
(b) Sound wave 2
(c) Sound wave 3
(d) They would all reach the listener at the same time

p. 330 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000327

Answer: Ans. (A) (d) Sound wave 4
Explanation: The sound wave with the highest frequency is sound wave 4 because it has the most compressed waves, indicating a higher frequency.
(B) (c) Sound wave 1 and sound wave 3
Explanation: The two sound waves that have almost the same loudness are sound wave 1 and sound wave 3, as they have similar amplitudes.
(C) (d) 35000 Hz
Explanation: The frequency of an ultras

source blocks: blk_000328

D. (D) (d) They would all reach the listener at the same time
Explanation: Since they are all sound waves traveling in the same medium (air), they travel at the same speed and would arrive simultaneously if emitted at the same time.

p. 331 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000333

Answer: Ans. (A) (d) Sound wave 4
Explanation: The sound wave with the highest frequency is sound wave 4 because it has the most

compressed waves, indicating a higher frequency.
(B) (c) Sound wave 1 and sound wave 3
Explanation: The two sound waves that have almost the same loudness are sound wave 1 and sound wave 3, as they have similar amplitudes.
(C) (d) 35000 Hz
Explanation: The frequency of an ultras

source blocks: blk_000328

18. 18. Sneha clapped her hand near a cliff and heard the echo after 4 s. (Speed of sound = 346 m/s)

p. 331 · numerical · solved_question · 4 marks (section_banner) · blocks: blk_000334

A. (A) What is the total distance covered by sound?
(a) 200 m (b) 250 m
(c) 1384 m (d) 692 m

p. 331 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000336

Answer: Ans. (A) (c) 1384 m
Explanation: Given,
Velocity of sound, v = 346 m/s
Time taken for hearing the echo, t = 4 s
Distance travelled= Velocity × Time
= 346 × 4
= 1384 m

source blocks: blk_000343

B. (B) Sound has to travel twice the distance between the cliff and Sneha.
(a) True
(b) False
(b) Cannot Say
(d) Sometimes travel twice and sometimes may travel thrice

p. 331 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000337

Answer: (B) (a) True
Explanation: Since the sound travels to the cliff and then reflects back, it covers twice the distance between Sneha and the cliff.

source blocks: blk_000344

C. (C) What is the distance of the cliff from Sneha?
(a) 1384 m (b) 692 m
(c) 348 m (d) 250 m

p. 331 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000338

Answer: (C) (b) 692 m
Explanation: The sound travels twice between the cliff and the individual in 4 s.
So, the distance between the individual and the cliff is = 1384 m / 2 = 692 m.

source blocks: blk_000346

D. (D) When we change feeble sound to loud sound, we increase its:
(a) amplitude (b) velocity
(c) frequency (d) wavelength

p. 331 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000339

Answer: (D) (a) amplitude
Explanation: The loudness or softness of a sound can be determined by its amplitude. A larger

amplitude means a louder sound.

source blocks: blk_000347

D. (D) Assertion (A): Echo is caused by the phenomenon of reflection.
Reason (R): The persistence of a sound in a big hall is due to echo.
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).

p. 331 · assertion_reason · solved_question · 4 marks (section_banner) · blocks: blk_000341

c. (c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

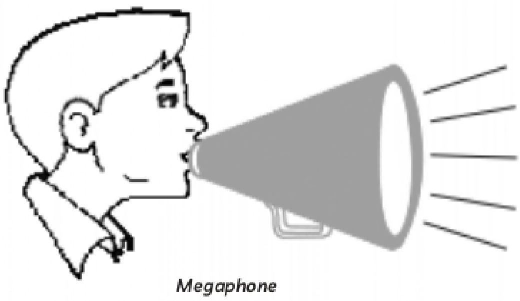
p. 331 · assertion_reason · solved_question · 4 marks (section_banner) · blocks: blk_000342

Answer: (C) (c) (A) is true but (R) is false.
Explanation: Sound waves do not require a polished surface for reflection like light waves. The surface for reflection of sound waves may be polished or rough.

source blocks: blk_000408

19. 19. Lata, while visiting London, attended an opera performance. Its architecture and furnishings appealed her. She observed that the draperies, cushions, and curtains on the curved ceiling were thoughtfully arranged. Additionally, she noticed a soundboard behind the stage. This made her curious whether these elements were solely for aesthetic purposes or had a scientific significance.

p. 331, 332 · case_study · solved_question · 4 marks (section_banner) · blocks: blk_000350



Megaphone

A. (A) What are the functions of curtains, pillows, and draperies in an opera house?

p. 331 · conceptual · solved_question · blocks: blk_000350

Answer: Ans. (A) Materials such as curtains, drapes, and cushions work as sound absorbent and decrease reverberation thereby, preventing excessive echoes in the hall.
(B) Soundboards and curved ceilings focus and direct sound so that it reaches all corners of the auditorium evenly. The curved design of the walls and ceiling also minimizes unwanted echoes and improves sound clarity by preventing multiple r

source blocks: blk_000357

B. (B) How do the curved ceiling and soundboard enhance the acoustics of the hall?

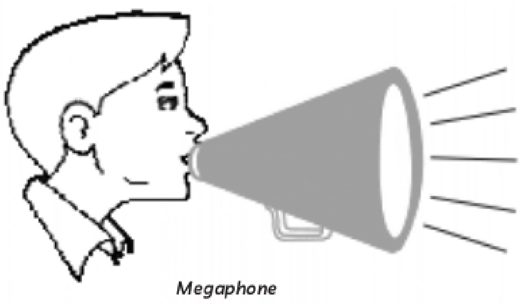
p. 331 · conceptual · solved_question · blocks: blk_000350

Answer: (B) (a) True
Explanation: Since the sound travels to the cliff and then reflects back, it covers twice the distance between Sneha and the cliff.

source blocks: blk_000344

C. (C) Loudhailers and horns are designed to project sound in a particular direction rather than spreading it in all directions as shown in the given figures. Explain the phenomenon.

p. 332 · long_answer · solved_question · 4 marks (section_banner) · blocks: blk_000352

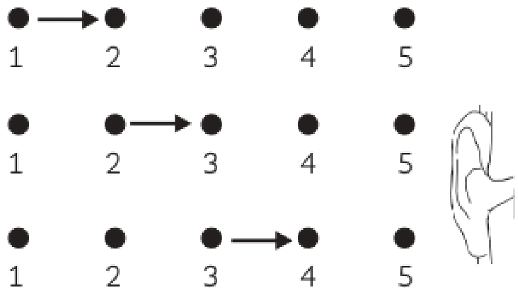


Answer: (C) In loudhailers and horns, In loudhailers and horns, the tube and conical opening reflect sound waves in a controlled manner, guiding them forward toward the audience instead of spreading in all directions.

source blocks: blk_000361

20. 20. Individual particles in a medium travel in a direction that is parallel to the disturbance's propagation direction. The particles do not travel from one location to another, instead oscillates back and forth around their resting position. Because sound waves spread in this manner, they are referred to as longitudinal waves. A transverse wave is another form of a wave. The particles of a transverse wave do not oscillate along the wave's propagation path, but rather up and down around their mean location as the wave travels. As a result, a transverse wave is one in which the medium's individual particles move around their mean positions in a direction perpendicular to the wave's propagation direction. Sound is conveyed through the air by compression waves, which rely on molecules transmitting energy to one another on a microscopic scale. Higher temperatures give air molecules more energy, causing them to vibrate more quicker. Because sound waves are pushed by collisions between molecules, they can travel faster as well.
(A) (i) Renu draws a figure to show the propagation of sound. Look at the figure carefully and explain how sound travels from the source to the listener.

p. 332 · long_answer · solved_question · 4 marks (section_banner) · blocks: blk_000362



ii. (ii) What other name is given to the sound waves?
(B) Give two characteristics of the medium that affect the speed of sound in it.
(C) What effect does the temperature have on the speed of sound in air?

p. 332 · conceptual · solved_question · 4 marks (section_banner) · blocks: blk_000364

Answer (partial): (B) The following are the two features of the medium that influence the speed of sound in it:
(1) Density: Sound generally travels slower in denser gases but faster in denser solids and liquids.
(2) Elasticity: A more elastic medium allows particles to return to their original positions more quickly, which increases the speed of sound.
(C) Temperature affects the speed of sound because at higher t

source blocks: blk_000367

21. 21. In the physics laboratory, Rishi conducted an experiment with a tuning fork. He observed that the object vibrated 1200 times per minute. Given that the speed of sound in air is 360 m/s, he proceeded with further calculations.

p. 333 · case_study · solved_question · 4 marks (section_banner) · blocks: blk_000368

A. (A) Find the frequency.
(a) 20 Hz (b) 50 Hz
(c) 100 Hz (d) 120 Hz

p. 333 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000369

Answer: Ans. (A) (a) 20 Hz
Explanation: Frequency = Number of vibrations = 1200 Hz
Time = 60 seconds
$$\text{Frequency} = \frac{\text{Number of vibrations}}{\text{Time in seconds}}$$
$$\nu = \frac{1200}{60} = 20\text{ Hz}$$

source blocks: blk_000376

B. (B) Find the wavelength.
(a) 12 m (b) 18 m
(c) 720 m (d) 7200 m

p. 333 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000370

Answer: (B) (b) 18 m
Explanation: Velocity of sound in air,
$$v = 360\text{ m/s}$$
$$\text{Velocity} = \text{Wavelength} \times \text{Frequency}$$
$$\lambda = \frac{360}{20}$$
$$\lambda = 18\text{ m}$$

source blocks: blk_000377

C. (C) What conclusion can be drawn from the fact that water splashes when the prongs of a sound-making tuning fork touch a beaker?
(a) The prongs of a tuning fork vibrate.
(b) The prongs of a tuning fork produce ultrasound.
(c) On the surface, waves are created.
(d) All of these

p. 333 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000371

Answer: (C) (a) The prongs of a tuning fork vibrate.
Explanation: When the vibrating prongs of a tuning fork touch the water surface, they transfer vibrations through the beaker into the water. These vibrations cause disturbances in the water, making it splash.

source blocks: blk_000378

D. (D) Which quantity remains the same in both the tuning fork's material as well as in the air when the tuning fork creates sound waves?
(a) Amplitude (b) Frequency
(c) Wavelength (d) All of these

p. 333 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000372

Answer: (D) (b) Frequency
Explanation: The frequency of a wave is determined by the vibrating source. It does not change when the medium changes. So, whether the sound travels through the tuning fork's material or through the air, its frequency remains constant.

source blocks: blk_000379

D. (D) Assertion(A): The water splashes when a tuning fork touches a beaker filled with water.
Reason(R): When the prong touches the beaker, light waves are produced on the surface and makes water come out.

p. 333 · assertion_reason · solved_question · 4 marks (section_banner) · blocks: blk_000374

a. (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

p. 333 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000375

Answer: (C) (c) (A) is true but (R) is false.
Explanation: Sound waves do not require a polished surface for reflection like light waves. The surface for reflection of sound waves may be polished or rough.

source blocks: blk_000408

22. 22. A ship on the surface of the water sends ultrasonic waves, which are reflected back by a submarine after 6 seconds. The speed of sound in water is 1450 meters per second.

p. 333, 334 · case_study · solved_question · 4 marks (section_banner) · blocks: blk_000382

A. (A) Distance travelled by sound in 6 seconds is:
(a) 362.5 m (b) 5.8 km
(c) 8.7 km (d) 10.8 km

p. 333 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000383

Answer: Ans. (A) (c) 8.7 km
Explanation: Given,
Velocity of sound = 1450 m/s
Time taken, t = 6 s
Distance travelled = Velocity of sound × Time taken
= 1450 × 6 = 8700 m
= 8.7 km

source blocks: blk_000390

B. (B) Distance of the submarine from the ship is:
(a) 2.9 km (b) 4.35 km
(c) 8.7 km (d) 9 km

p. 333 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000384

Answer: (B) (b) 4.35 km
Explanation: The minimum distance between a sound source (ship) and a reflector (submarine) is twice that of reflected sound. So, for the reflected sound or signal to be heard well, the submarine must be at least 4350 metres or 4.35 kilometres away from the ship.
Distance of submarine = Total distance / 2
= 8700 m / 2 = 4350 m = 4.35 km

source blocks: blk_000391

C. (C) Assertion(A): Bullets and jet crafts travel at supersonic speeds.
Reason(R): Bullets and jet crafts move faster than the speed of sound.
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

p. 334 · assertion_reason · solved_question · 4 marks (section_banner) · blocks: blk_000386

Answer: (C) (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
Explanation: Bullets and jet crafts travel at supersonic speeds because they move faster than the speed of sound.

source blocks: blk_000392

D. (D) The effect of any sound lasts roughly 0.1 seconds in our ear. This is referred to as:
(a) Resistance of hearing
(b) Persistence of hearing
(c) Echo
(d) Frequency

p. 334 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000387

Answer: (D) (b) Persistence of hearing
Explanation: It is the sensation of hearing that persists in the human ear. This sensation persists for 1/10 th of a second.

source blocks: blk_000393

D. (D) How is the time period (T) related to the frequency (ν) of a wave? Identify the expression.
(a) $T = 1/\nu$
(b) $T = \nu$
(c) $T = 1/\nu^2$
(d) $T \neq 1/\nu$

p. 334 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000389

Answer: (D) (a) $T = 1/\nu$
Explanation: Time period = 1 / Frequency

source blocks: blk_000395

23. 23. Sachin's mother is pregnant. Doctor suggested to perform ultrasonography to check the health of baby. She wondered about the technique used for the purpose and asked the lab attendant some questions.

p. 334 · case_study · solved_question · 4 marks (section_banner) · blocks: blk_000396

A. (A) The frequency of waves used in ultra-sonography is:
(a) 2 Hz to 20 Hz
(b) 20 Hz to 20000 Hz
(c) 2000 Hz to 20000 Hz
(d) 2 MHz to 18 MHz

p. 334 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000398

Answer: Ans. (A) (d) 2 MHz to 18 MHz
Explanation: The word "ultrasound" in Physics refers to all acoustic energy with a frequency higher than human hearing (above 20,000 hertz or 20 kilohertz). Diagnostic sonographic scanners typically operate at a frequency of 2 to 18 megahertz, which is hundreds of times higher than the human hearing limit.

source blocks: blk_000405

B. (B) Ultrasonic sound has .... frequency.
(a) above 20 kHz
(b) below 20 kHz
(c) between 2-20 kHz
(d) none of the above

p. 334 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000399

Answer: (B) (a) above 20 kHz
Explanation: Ultrasonic sound has a frequency range above 20 kHz.

source blocks: blk_000406

C. (C) Assertion(A): Sound waves get reflected when they fall on the surface of an obstacle.
Reason(R): Sound waves also need a polished surface for reflection like light waves.
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

p. 334 · assertion_reason · solved_question · 4 marks (section_banner) · blocks: blk_000400

Answer: (C) (c) (A) is true but (R) is false.
Explanation: Sound waves do not require a polished surface for reflection like light waves. The surface for reflection of sound waves may be polished or rough.

source blocks: blk_000408

D. (D) Choose the principle of ultrasound from the below options on which it works:
(a) Refraction of sound waves
(b) Reflection of sound waves
(c) Propagation of sound waves
(d) Both (a) and (b)

p. 334 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000401

Answer: (D) (d) Both (a) and (b)
Explanation: The ultrasound works on the principle of reflection and refraction. While it is necessary that sound waves need a medium to travel, so we can say propagation is important but it is only because the sound wave gets refracted when the medium changes and are reflected back that the image is formed. Thus, the principle for ultrasound is reflection and refraction.

source blocks: blk_000409

D. (D) Identify the applications of ultrasound.
(a) Detection of cracks
(b) Echocardiography
(c) Lithotripsy
(d) All of the above

p. 334 · mcq · solved_question · 4 marks (section_banner) · blocks: blk_000403

Answer: (D) (d) All of the above
Explanation: Ultrasound is used in multiple applications, including:
(1) Detection of cracks in metal structures.
(2) Echocardiography (heart imaging).
(3) Lithotripsy (breaking kidney stones using ultrasound waves).

source blocks: blk_000411

24. 24. How can we recognise our friends from their voices? [DIKSHA]

p. 335 · short_answer · solved_question · 2 marks (section_banner) · blocks: blk_000414

Answer: Ans. Timbre and pitch are key characteristics of sound that help distinguish different voices. Due to variations in timbre and pitch, a person can recognise a friend's voice even in a dark room where visual identification is not possible.

source blocks: blk_000415

25. 25. If any explosion takes place at the bottom of a lake, what type of shock waves in water will take place? [NCERT Exemplar]

p. 335 · conceptual · solved_question · 2 marks (section_banner) · blocks: blk_000416

Answer: Ans. When an explosion occurs at the bottom of a lake, longitudinal waves are generated in the water. In longitudinal waves, the particles of the medium oscillate parallel to the direction of wave propagation. These waves travel through the water as compressions and rarefactions, similar to how sound waves propagate through air.

source blocks: blk_000417

26. 26. Define frequency and write its SI unit.

p. 335 · short_answer · solved_question · 2 marks (section_banner) · blocks: blk_000419

Answer: Ans. The frequency of a sound wave refers to the number of oscillations produced per unit time. Hertz (Hz) is the SI unit of frequency.

source blocks: blk_000420

27. 27. What is a transverse wave, and how does it work? Is it possible to produce it in any medium: solid, liquid, or gas?

p. 335 · short_answer · solved_question · 2 marks (section_banner) · blocks: blk_000423

Answer: Ans. A transverse wave is a type of wave in which the particles of the medium oscillate perpendicular to the direction of wave propagation.
Transverse waves can propagate through solids and on the surface of liquids, but they cannot travel through liquids or gases because these states of matter do not provide the necessary restoring forces for perpendicular oscillations.

source blocks: blk_000424

28. 28. The sound of thunder is preceded by a flash of lightning. Justify

p. 335 · conceptual · solved_question · 2 marks (section_banner) · blocks: blk_000425

Answer: Ans. This is due to the fact that light travels much faster than sound.
The speed of light in air is approximately 3×10^8 m/s, while the speed of sound in air is about 340 m/s.

source blocks: blk_000426

29. 29. Is there any relation among the wavelength? (λ) of a sound wave, speed (v) and frequency (ν)? If yes, write the relationship.

p. 336 · numerical · solved_question · 2 marks (section_banner) · blocks: blk_000428

Answer: Ans. Yes, there is a relation among the above-mentioned terms, i.e.,
$$\lambda = \frac{v}{\nu}$$
∴ Wavelength = Speed or Velocity / Frequency

source blocks: blk_000429

30. 30. What are the applications of ultrasonic waves in medicine? [DIKSHA]

p. 336 · short_answer · solved_question · 2 marks (section_banner) · blocks: blk_000430

Answer: Ans. Ultrasonic waves have several important applications in medicine:
(1) Echocardiography: Ultrasonic waves are used to reflect from various parts of the heart, forming detailed images that help diagnose heart conditions.
(2) Ultrasound Scanning: Ultrasound scanners use ultrasonic waves to create images of internal organs, helping in diagnosing diseases and monitoring pregnancies.
(3) Lithotrips

source blocks: blk_000431

31. 31. Dolphins can locate their prey underwater by using sound waves. They release sound waves that travel, hit the prey and reflect to them.

p. 336 · case_study · solved_question · 2 marks (section_banner) · blocks: blk_000432

A. (A) What is the primary method dolphins use to locate prey underwater?
(B) What do the sound waves released by dolphins do when they encounter prey?

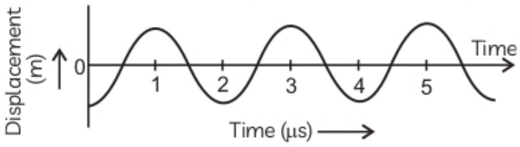
p. 336 · short_answer · solved_question · 2 marks (section_banner) · blocks: blk_000434

Answer: Ans. (A) The primary method dolphins use to locate prey underwater is echolocation.
(B) The sound waves released by dolphins travel, hit the prey, and reflect (or bounce) back to the dolphin.

source blocks: blk_000437

32. 32. The given graph shows the displacement versus time relation for a disturbance travelling with a velocity of 1500 m/s. Calculate the wavelength of the disturbance.

p. 336 · numerical · solved_question · 2 marks (section_banner) · blocks: blk_000438



Answer: Ans. Velocity, v = 1500 m/s
From given graph, time period for one wave,
$$t = 2\text{ }\mu\text{s} = 2 \times 10^{-6}\text{ s}$$
$$\text{Velocity }(v) = \frac{\text{Wavelength }(\lambda)}{\text{Time }(t)}$$
$$\lambda = vt$$
$$\lambda = 1500 \times 2 \times 10^{-6}$$
$$\lambda = 3 \times 10^{-3}$$
Hence, the wavelength of the disturbance is $3 \times 10^{-3}\text{ m}$.

source blocks: blk_000441

33. 33. Define sound. How is it produced?

p. 336 · short_answer · solved_question · 2 marks (section_banner) · blocks: blk_000442

Answer: Ans. A sound is a kind of energy that causes our ears to vibrate and produce the feeling of hearing. When an object vibrates or moves in a 'to-and-fro' motion, a sound is created.
For example, When a string is stretched by holding one end between the teeth and the other end in one hand, plucking the string in the middle with the other hand generates a vibration, which causes the sound to be create

source blocks: blk_000443

34. 34. While playing cricket, Ratan told his friend that he saw the ball hit the bat before hearing the sound of the impact. Why did this happen? Support Ratan's observation with suitable reason.

p. 336 · short_answer · solved_question · 2 marks (section_banner) · blocks: blk_000444

Answer: Ans. The ball is seen to hit the bat first, and the sound of hitting is heard a little later because light travels much faster than sound. The speed of light in air is approximately 3×10^8 m/s, whereas the speed of sound is around 343 m/s.

source blocks: blk_000445

35. 35. Which of the following travels fastest and slowest in the air? Light, Sound and Supersonic Aircraft

p. 336 · conceptual · solved_question · 2 marks (section_banner) · blocks: blk_000446

Answer: Ans. Light travels fastest in air with a speed of 3×10^8 m/s because light waves do not need a medium to travel. Supersonic aircraft's speed is greater than the speed of sound. Sound is the slowest of all.

source blocks: blk_000447

36. 36. Two colleagues were playing identical guitars that were tuned to the same pitch. Will the frequency of notes be of the same high quality? Explain your answer.

p. 337 · short_answer · solved_question · 2 marks (section_banner) · blocks: blk_000450

Answer: Ans. The two notes do not necessarily have the same quality. While the frequency of a note determines its pitch, its quality (or timbre) is influenced by the harmonic or overtone content. The quality of a sound depends on the number, distribution, and relative intensity of these harmonics and overtones. Even if two notes have the same pitch, their timbre may still differ due to variations in these

source blocks: blk_000451, blk_000452

37. 37. Look at the arrangement of glass bottles and answer the questions that follow:

p. 337 · case_study · solved_question · 3 marks (section_banner) · blocks: blk_000455

A. (A) Which bottle will produce the highest pitch if we tap the bottle with a spoon? Which bottle will produce the lowest pitch?

p. 337 · conceptual · solved_question · blocks: blk_000455

Answer: Ans. (A) When we tap the bottles with a spoon one by one, bottle 'a' will produce the highest pitch and bottle 'h' will produce the lowest pitch. When we hit the bottle with the spoon, the glass vibrates and it is these vibrations, that ultimately produce the sound. It is seen that tapping an empty bottle or bottle with less water produces a higher-pitched sound than tapping a bottle almost full o

source blocks: blk_000457

B. (B) How do the bottles differ between the highest pitch to the lowest pitch?

p. 337 · conceptual · solved_question · blocks: blk_000455

Answer: Adding water to the bottle dampens the vibrations created by stirring the glass with a spoon. Less water allows the glass to vibrate faster, producing a higher pitch. Conversely, adding more water slows the vibrations, resulting in a lower pitch.
(C) The same bottle 'a' that produces a high-pitched sound when tapped with a spoon creates a low-pitched sound when blown across the top. This differenc

source blocks: blk_000459

C. (C) Why does blowing across the top of the bottle and tapping the bottle produce different sounds?

p. 337 · conceptual · solved_question · blocks: blk_000455

Answer (partial): larger volume of air is available to vibrate. When water is added, the air space inside decreases, meaning less air is available to vibrate. This causes the vibrations to happen more quickly, producing a higher-pitched sound.

source blocks: blk_000460

38. 38. A girl is sitting in the middle of a park of dimension 12 m × 12 m. On the left side of it, there is a building adjoining the park and on the right side of the park, there is a road adjoining the park. A sound is produced on the road by a cracker. Is it possible for the girl to hear the echo of this sound? Explain your answer. [NCERT Exemplar]

p. 337 · numerical · solved_question · 3 marks (section_banner) · blocks: blk_000461

Answer: Ans. An echo can only be heard if the time difference between the original sound and the reflected sound received by the receiver is less than 0.1 seconds.
The minimum distance travelled by the reflected sound wave for clearly hearing the echo is: Distance = Velocity of sound × Time interval
= 344 × 0.1
= 344.4 m. However, in this situation, the sound reflected from the building and subsequently r

source blocks: blk_000462, blk_000463, blk_000464

39. 39. For studying the reflection of sound, why is it important to have a smooth reflecting surface? [DIKSHA]

p. 337 · long_answer · solved_question · 3 marks (section_banner) · blocks: blk_000465

Answer: Ans. A smooth reflecting surface is important for studying the reflection of sound because it ensures that the sound waves follow the laws of reflection—where the angle of incidence equals the angle of reflection.
If the surface is rough, the sound waves scatter in different directions instead of reflecting uniformly, making it difficult to observe and study sound reflection properly. Smooth surfa

source blocks: blk_000466

40. 40. How do the following factors affect the speed of sound in air?
(A) Air frequency
(B) Air temperature
(C) Air pressure
(D) Moisture

p. 338 · long_answer · solved_question · 3 marks (section_banner) · blocks: blk_000472

Answer: Ans. The following factors have an impact on sound speed in the following ways:
(A) Air frequency: The speed of sound is not affected by frequency. It remains constant for a given medium and temperature.
(B) Air temperature: As temperature increases, the speed of sound also increases because warmer air has more energetic molecules, allowing sound waves to propagate faster.
(C) Air pressure: The sp

source blocks: blk_000473

41. 41. Vivaan was positioned in line with two tanks and observed them firing at each other simultaneously. However, he heard the sounds of the two shots 2 seconds and 3.5 seconds after seeing the flashes. If the distance between the two tanks is 510 meters, determine the speed of sound.

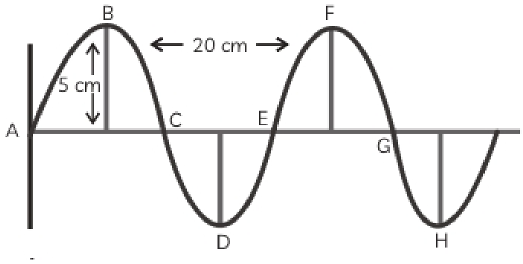
p. 338 · numerical · solved_question · 3 marks (section_banner) · blocks: blk_000474

Answer: Ans. The time taken to hear the first shot, t_1 = 3.5 s
The time taken to hear the second shot, t_2 = 2 s
Distance between the two tanks = 510 m
The difference in time between the two sounds is:
= 3.5 - 2 s = 1.5 s
Since the sound travels the entire distance of 510 m in 1.5 s, the speed of sound:
Speed = Distance / Time
Speed = 510 / 1.5 = 340 m/s
Hence, the speed of sound for a distance of 510 m be

source blocks: blk_000475

42. 42. Waves of frequency 100 Hz are produced in a string as shown in figure.

p. 338 · numerical · solved_question · 3 marks (section_banner) · blocks: blk_000476



Answer: Ans. (A) The amplitude of a wave is its maximum displacement.
The wave's amplitude from the given figure is 5 cm.
(B) The wavelength of a wave is the distance between two consecutive crests and troughs.
The wave's wavelength, as shown in the diagram, is 20 cm (0.2 m).
(C) A wave's velocity is given by:
Velocity = Wavelength × Frequency
Frequency of wave = 100 Hz (given)
v = 0.2 × 100 = 20 m/s

source blocks: blk_000479

53. Give its:
(A) Amplitude
(B) Wavelength
(C) Velocity [NCERT Exemplar]

p. 338 · numerical · in_text · 3 marks (section_banner) · blocks: blk_000478

Answer: Ans. (A) The amplitude of a wave is its maximum displacement.
The wave's amplitude from the given figure is 5 cm.
(B) The wavelength of a wave is the distance between two consecutive crests and troughs.
The wave's wavelength, as shown in the diagram, is 20 cm (0.2 m).
(C) A wave's velocity is given by:
Velocity = Wavelength × Frequency
Frequency of wave = 100 Hz (given)
v = 0.2 × 100 = 20 m/s

source blocks: blk_000479

43. 43. Represent graphically by two separate diagrams in each case.

p. 338, 339 · conceptual · solved_question · 5 marks (section_banner) · blocks: blk_000481

A. (A) Two sound waves having the same amplitude but different frequencies.
(B) Two sound waves having the same frequency but different amplitudes.

p. 338 · conceptual · solved_question · blocks: blk_000481

Answer (partial): Ans. (A) Two sound waves having the same amplitude but different frequencies.

source blocks: blk_000483

B. (B) Two sound waves having the same frequency but different amplitudes.

p. 339 · conceptual · solved_question · 5 marks (section_banner) · blocks: blk_000487

Answer: (B) Echocardiography: Another popular medical application of ultrasonography is echocardiography, sometimes known as ECG. Again, this is an imaging technology in which sound waves pass through the body to produce an image of a patient's cardiac condition.
Ultrasonic waves are utilised in electrocardiography to build an image of the heart via reflection and

detection of these waves from various are

source blocks: blk_000501

C. (C) Two sound waves having different amplitudes and also different wavelengths. [NCERT Exemplar]

p. 338 · conceptual · solved_question · 5 marks (section_banner) · blocks: blk_000482

Answer: Ans. (A) Two sound waves having the same amplitude but different frequencies.

source blocks: blk_000483

C. (C) Two sound waves having different amplitudes and also different wavelengths.

p. 339 · conceptual · solved_question · 5 marks (section_banner) · blocks: blk_000490

NOT IN DOCUMENT — no answer present in the source

44. 44. (A) What is meant by the reflection of sound waves? [DIKSHA]
(B) Describe an activity to study the reflection of sound.

p. 339 · long_answer · solved_question · 5 marks (section_banner) · blocks: blk_000493

Answer: Ans. (A) Reflection of sound waves refers to the phenomenon where sound waves bounce back into the same medium after striking a surface of another medium, following the laws of reflection.
(B) Take two identical pipes. Arrange them on a table near a wall. Keep a clock near the open end of one of the pipes and try to hear the sound of the clock through the other pipe. Adjust the position of the pip

source blocks: blk_000494, blk_000495

45. 45. Define:
(A) Echolocation
(B) Echocardiography
(C) Ultrasonography

p. 339 · long_answer · solved_question · 5 marks (section_banner) · blocks: blk_000497

Answer: Ans. (A) Echolocation: The use of sound waves and echoes to locate things in space is known as echolocation. Bats use echolocation to explore and find their meal in the dark.
They also use echolocation to travel and discover insect meals. Ultrasound is a term for sound waves that are produced at frequencies higher than those heard by humans. Bats' sound waves bounce off the items in their environm

source blocks: blk_000498

46. 46. A stone is thrown from the top of an 8000 metre high tower into a water pond at the base of the tower. When can you hear the splash from the top? Given, $g = 10\ ms^{-2}$ and sound speed = 340 m/s.

p. 340 · numerical · solved_question · 5 marks (section_banner) · blocks: blk_000505

Answer (partial): Ans. Height of the tower, s = 8000 m
Velocity of sound, v = 340 m/s
Acceleration due to gravity, $g = 10\ m/s^2$
Initial velocity of the stone, u = 0 (since the stone is initially at rest)
Time taken by the stone to fall to the base of the tower, t_1
According to the second equation of motion: $s = ut_1 + 1/2\ gt_1{}^2$ $8000 = 0 \times t_1 + 1/2\ (10)\ t_1{}^2$ $8000 = 1/2\ (10)\ t_1{}^2$ $t_1{}^2 = 1600$

source blocks: blk_000506, blk_000507, blk_000508, blk_000509

47. 47.

p. 340 · unknown · solved_question · 5 marks (section_banner) · blocks: blk_000516

A. (A) Draw a curve showing density or pressure variations with respect to distance for a disturbance produced by sound. Mark the position of compression and rarefaction on this curve. Also, define wavelengths and time period using this curve.

p. 340 · conceptual · solved_question · blocks: blk_000516

Answer (partial): Ans. (A) Curve showing density or pressure variations with respect to distance for a disturbance produced by sound: Wavelength is the distance between two consecutive compressions or two consecutive rarefactions.
Time period is the time taken to travel the distance between any two consecutive compressions or rarefactions from a fixed point.
(B) (i) Amplitude is 5 cm.
Amplitude is the maximum displ

source blocks: blk_000520, blk_000522

B. (B) Waves of frequency 100 Hz are produced in a string as shown in figure. Give its (i) amplitude, (ii) wavelength.

p. 340 · numerical · solved_question · 5 marks (section_banner) · blocks: blk_000518

Answer: Wavelength is the distance between two consecutive compressions or two consecutive rarefactions.
Time period is the time taken to travel the distance between any two consecutive compressions or rarefactions from a fixed point.
(B) (i) Amplitude is 5 cm.
Amplitude is the maximum displacement from the mean position i.e., BM.
(ii) Wavelength is 20 cm.
Distance between two crests or two trough i.e., B

source blocks: blk_000522

1. 1. The least distance, a person must be from a sound-reflecting surface to hear an echo at 35 degrees Celsius is:
(a) 34.4 m
(b) 68.8 m
(c) 8.6 m
(d) 17.2 m

p. 341 · mcq · self_assessment · 1 mark (explicit) · blocks: blk_000527

NOT IN DOCUMENT — no answer present in the source

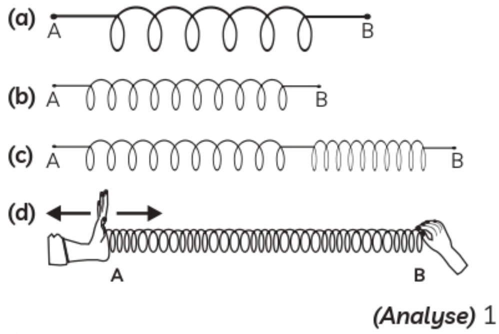
2. 2. Which one of these depends on the intensity of a sound?
(I) speed of the sound
(II) loudness of the sound
(III) frequency of the sound
Options:
(a) Only (I)
(b) Only (II)
(c) Only (III)
(d) Both (II) and (III)

p. 341 · mcq · self_assessment · 1 mark (explicit) · blocks: blk_000528

NOT IN DOCUMENT — no answer present in the source

3. 3. Four students placed slinkies in a horizontal position, securing end B while keeping a fixed distance between the loops without any initial compression or stretching.
When they pushed and pulled end A, a series of compressions and rarefactions were generated. Each student then sketched a diagram to represent the wave pattern formed in the slinky. Which of their diagrams accurately illustrates the correct formation of compressions and rarefactions?

p. 341 · conceptual · self_assessment · 1 mark (section_banner) · blocks: blk_000529



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4. 4. Radha writes several statements related to sound terminology, but two of them are incorrect. Identify the incorrect statements:
(I) A sound of single frequency is called a note.
(II) Pitch helps in differentiating between a shrill sound from a flat sound.
(III) A sound which is produced due to a mixture of several frequencies is called a tone.
(IV) A sound which has a pleasing effect on the listener is called music.
Options:
(a) (I) and (II)
(b) (I) and (III)
(c) (I) and (IV)
(d) (II) and (IV)

p. 341 · mcq · self_assessment · 1 mark (explicit) · blocks: blk_000532

NOT IN DOCUMENT — no answer present in the source

5. 5. Assertion(A): Reverberation is desirable.
Reason(R): Reverberation can be reduced by increasing the absorption of sound energy.

p. 341 · assertion_reason · self_assessment · 1 mark (explicit) · blocks: blk_000535

NOT IN DOCUMENT — no answer present in the source

6. 6. Assertion(A): The sound travelling through rails is louder than the sound through air.
Reason(R): Intensity of sound is directly proportional to the density of the medium.

p. 341 · assertion_reason · self_assessment · 1 mark (explicit) · blocks: blk_000536

NOT IN DOCUMENT — no answer present in the source

7. 7. Read the given extract and answer the following questions.
Thunder and lighting are natural phenomena. While Meenal was playing in the park, she saw the dark clouds gathering in the sky followed by thunder and lightning. She got scared and quickly ran home as heavy rain began to fall. Curious about the event, she asked her mother some questions.

p. 341 · case_study · self_assessment · 1 mark (explicit) · blocks: blk_000538

A. (A) Why is thunder followed by lightning?

p. 341 · conceptual · self_assessment · blocks: blk_000538

NOT IN DOCUMENT — no answer present in the source

B. (B) Sound produced by a thunderstorm is heard 10 s after the lightning is seen. Calculate the approximate distance of the thunder cloud.

p. 341 · numerical · self_assessment · 1 mark (explicit) · blocks: blk_000539

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C. (C) If velocity of sound in air is 340 ms^{-1}.
Calculate:
(i) Frequency when wavelength is 1.2 m.
(ii) Wavelength whose frequency is 265 Hz.

p. 341 · numerical · self_assessment · 1 mark (section_banner) · blocks: blk_000540

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C. (C) Meenal saw thunder and lightning while playing outside. What safety measures would you take during a thunderstorm

p. 341 · short_answer · self_assessment · 1 mark (section_banner) · blocks: blk_000542

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66. to protect yourself, and why? Explain the science behind it. (Apply) 2

p. 342 · short_answer · self_assessment · 2 marks (explicit) · blocks: blk_000544

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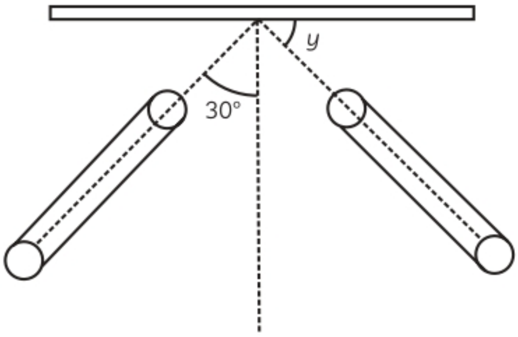
8. 8.

p. 342 · unknown · self_assessment · 2 marks (explicit) · blocks: blk_000546



A. (A) How will you differentiate between roar of tiger and buzzing of mosquito?

p. 342 · distinguish · self_assessment · blocks: blk_000546

NOT IN DOCUMENT — no answer present in the source

B. (B) In the following figure, determine the angle of reflection and value of y.
(Apply) 2

p. 342 · numerical · self_assessment · blocks: blk_000546

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9. 9. Among the two given curves (a) and (b) in the figure, which one is more likely to represent a male voice? Give a reason for your answer.
(Apply) 2

p. 342 · conceptual · self_assessment · 2 marks (explicit) · blocks: blk_000548

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10. 10. Give reasons for the following:

p. 342 · long_answer · self_assessment · 3 marks (explicit) · blocks: blk_000551

A. (A) A reverberation time of a hall used for speeches should be very short.

p. 342 · conceptual · self_assessment · blocks: blk_000551

NOT IN DOCUMENT — no answer present in the source

B. (B) Why a sound board used in the hall have a curved surface?

p. 342 · conceptual · self_assessment · blocks: blk_000551

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C. (C) A man puts his ear against one end of a rail while another man strikes the opposite end with a metal hammer. As a result, the man hears two distinct sounds separated by a short interval.
(Apply) 3

p. 342 · conceptual · self_assessment · blocks: blk_000551

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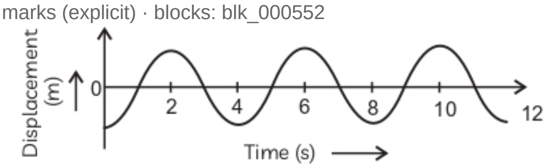
11. 11. The given graph shows the displacement verses time relation for a disturbance travelling with a velocity of 2500 m/s.

p. 342 · numerical · self_assessment · 3 marks (explicit) · blocks: blk_000552



A. (A) Look at the graph carefully and calculate:
(i) time period
(ii) frequency
(iii) wavelength of the disturbance
(Apply)

p. 342 · mcq · self_assessment · blocks: blk_000552

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B. (B) Define oscillation and amplitude in context of sound. (Understand) 3

p. 342 · conceptual · self_assessment · blocks: blk_000552

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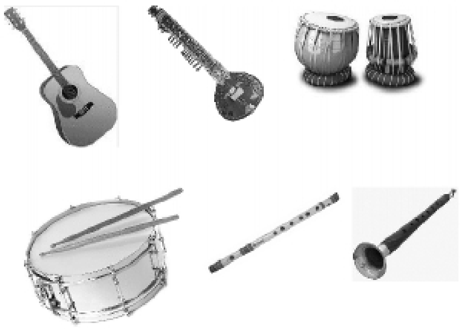
12. 12. Seema enjoys listening to instrumental music and has purchased some musical instruments to play. She observed that different musical instruments produce distinct sounds.

p. 342 · case_study · self_assessment · 5 marks (section_banner) · blocks: blk_000555

A. (A) Do these sounds reach us with same speed or different speed? Explain your answer with a reason.
(B) Seema observed a guitarist plucking a guitar string. What types of waves are produced in:
(i) air?
(ii) water?
Provide reasons to support your answer.
(C) Before playing the orchestra in a musical concert, the sitarist tries to adjust the tension and pluck the string suitably. By doing so, what is he trying to adjust?
(D) Identify which part vibrates to produce sound in the following musical instruments: Tabla, Drum, Flute, Mouth organ, Sitar, Piano, Violin, Harmonium and Shehnai. (Understand) 5

p. 342 · long_answer · self_assessment · 5 marks (explicit) · blocks: blk_000557



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